Advanced Empirical Research Method
Program Evaluation with Observational Data
Chapter 1 - Concepts, Definitions and Controversy

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Course objectives

Motivation

Concepts of Causal Inference

Bounds

Spillovers and Experiments: Based on slides by Francis DiTraglia

A controversy
- “The credibility revolution...”
- “A False panacea...”

Course Objectives

This part of the course is in continuation of the econometrics methods for causal inference that have been introduced in Core ERM.

Within the framework of causal inference (or more specifically program evaluation), we will discuss in greater length methods for addressing the so-called **fundamental problem of causal inference**:

1. Selection-on-observables;
2. Instrumental variable methods (LATE Model, Roy model, Marginal Treatment Effect);
3. Difference-in-difference;
4. Regression Discontinuity Design.

Course Objectives

Core ERM made sure you understood the basic intuitions of each method.
But the basic intuition is often insufficient when you face a concrete problem.

Take the question of measuring the employment effects of the introduction of the minimum wage in a US state. The basic idea of a Diff-in-Diff is to find a counterfactual state with a parallel trend and measure departure from this trend. This leaves the **practical questions**:

- How do I check that trends across regions are indeed parallel?
- How do I deal with regional and time-varying confounders?
- What if I have multiple regions and the treatment time differ across regions?
- State-level employment rates are correlated over time, how does it affect inference?

Course Objectives

The idea of this course is to go beyond the basic intuition and discuss the issues in their practical implementation. Such a discussion cannot be exhaustive:

- first, because of the time constraint,
- second, because there is a great number of ways to add sophistication to a basic intuition.

My objectives are to:

1. present **common issues** that a researcher faces when implementing causal inference methods, that is, questions you should ask yourself when using those methods;
2. share **standard practices** to address those issues, for example when assessing the plausibility of an assumption;
3. *introduce* you to relevant R-packages that implement these standard methods.

Course Objectives

The approach of each lecture:

1. Quick recap of the basic principles, assumptions and identification results:
   My assumption is that you are familiar with the basic intuition. If you are not yet confident with the material of Core ERM, please review the lecture notes:
   https://www.treatment-effects.com/treatment-effects.pdf.
2. Present **standard** issues and **standard** data analyses associated with each method.
3. Point you to **one of possibly many** R-packages that implements these analyses.
4. Point you to **possible extensions** of the method.

Course Objectives

Disclaimer:

- Most of you are no doubt better programmers and R-experts than I am, and there will surely be ways to optimise the codes I will share. It would be extremely helpful if you would share suggestions with the class.

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What do we mean with ‘program evaluation’ ?

Motivation

Causal inference is often part of a larger endeavour called textcolorblueprogram evaluation. The goals are:

1. Monitor the implementation of public or firm policies
2. Evaluate the impact of these policies on the well-being of its beneficiaries
3. Evaluate possible extensions to the whole population

We will see this with a specific example.

- Cash transfers conditional on health and education:
  these have been implemented in Mexico, Brazil, Colombia, etc. 90s and 2000s.

Motivation - Example of PROGRESA

Example of PROGRESA or Oportunidades in Mexico from 1997 to today.

Policy objectives: reduce child work and poverty.

- In the short run, by providing conditional cash transfers.
- In the long run, by increasing education level.

Economic arguments:

- Poor households are liquidity constrained,
- They do not invest optimally in education and health,
- Good education and health encourage growth, alleviate poverty and increase well-being in the long run.

Motivation - Example of PROGRESA

Target of policy: Poor families with children of schooling age. Poverty defined according to income, level of education, characteristics of housing, number of persons in the household.

Treatment: Conditional cash transfers:

- on average, 22% of family income;
- subsidies increase with grade level and premium for girls enrolled in lower secondary education;
- transfers represent two-third of average full-time earnings for a working child.

Requirement: Minimum child school attendance and regular visits to the doctor.

Motivation - Example of PROGRESA

Timing and scope: Policy put in effect slowly by villages starting in summer 98 and then extended to more villages. First implemented in rural areas, later extended to the whole country. In 98 and 99, aid to 2.6 millions poor families (40% of rural families).

Early implementation phase: In 97 and 98, 500 villages are split into two categories:

- the one where policy is effective (treatment group) and poor households are treated;
- other villages (control group) where the policy is not introduced.

Data for quantitative evaluation: household surveys on participant AND non-participants. Two baseline surveys before the implementation and two follow-up surveys a year apart.

Motivation - Example of PROGRESA

Quantitative Evaluation:

1. Monitoring: Do transfers go to poor families?

- Compare incidence of the program by income deciles.

2. Evaluation of Impact: What is the impact of the policy on school attendance, education level, school performance, etc.? Are beneficiaries better off than if there would have been **no program?**

- Compare outcome for beneficiaries and non-beneficiaries. The construction and interpretation of the evaluation method depends on whether the data are experimental or observational.

Motivation - Example of PROGRESA

Quantitative Evaluation (continued):

3. Cost and benefits analysis: What is the net cost of the program? textcolorblueIs the program sustainable?

- Cost: Aggregate monetary transfers to poor households
- Benefits: long-run earnings and health
- account for **opportunity costs**, e.g., for households since children are not able to work on the farm or family business.

Motivation - Example of PROGRESA

Evaluation methods with observational data are the topic of this course. It is important to keep in mind that the goals of a program evaluation are larger.

The **researcher** might be curious about the causal effect of some treatment D on an outcome Y. But the **policy maker** is ultimately interested in the question "what is the cheapest way of improving/reducing Y?"

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Causality

Questions about **causal effect** of a program or **treatment** D on an **outcome** Y.

"Treatment" (biostatistics / medical trials jargon): any variable D whose causal effect we hope to learn.

For learning causal effects, randomization is seen by many social scientists as the "golden standard". But this special status of randomization in social science is challenged.

- Would behavior be the same as when access is granted randomly? Note the '*Consistency condition*': choosing treatment is the same as being allocated.
- What do we do if we cannot randomize for biological, political or ethical reasons?
- What if there is **non-compliance** in experimental settings? Treatment receipt is no longer randomly assigned, even if treatment offer is.

The Fundamental Problem of Causal Inference

The fundamental problem of causal inference:
We can never observe a person's **counterfactual** outcome.

Counterfactual = **within** person/unit comparison.

Key idea: Find in the population a similar person/unit that has not received the treatment.

The Fundamental Problem of Causal Inference

The methods we will see:

1. **Selection on observable:** similar individuals/units are defined by their observable characteristics.
2. **Difference-in-difference:** similar individuals/units shared the same trend as the treated group before the program implementation.
3. **Regression Discontinuity Design:** Similar individuals/units are those very close to a defined cut-off, but have not been taken in the program.
4. **Instrumental Variables:** Similar individuals are those who have not entered the program because of an exogenous/orthogonal shock.

Potential outcomes framework

Potential outcomes framework, a.k.a. the Rubin Causal Model (RCM).
Each person i, a pair of **potential outcomes** (y_{i0}, y_{i1}) and treatment status d_i.

Example of return to graduate education

- y_{i0} lifetime earning without PROGRESA,
- y_{i1} lifetime earning with PROGRESA,

We observe:

$$y_i = (1 - d_i)y_{i0} + d_i y_{i1} = y_{i0} + d_i(y_{i1} - y_{i0}) \tag{1}$$

Assumption (stable unit treatment value assumption (SUTVA))

i's outcome depends only on her own treatment and not the treatments of anyone else.

Potential Outcome Framework

Accompanying the definition of potential outcomes, definition of:

- population of interest,
- individual characteristics $(y_{0i}, y_{1i}, d_i, \boldsymbol{w}_i)$,
- Sample of the population of size n,
- Random variables $(Y_0, Y_1, D, \boldsymbol{W})$, where randomness comes from observing a random sample of the population.

$$Y = (1 - D)Y_0 + DY_1 = Y_0 + D(Y_1 - Y_0). \tag{2}$$

Knowledge of the joint distribution of $(Y_0, Y_1, D, \boldsymbol{W})$ implies knowledge of the joint distribution of $(Y, D, \boldsymbol{W})$. But it is the converse we are interested in.

Population parameters of interest

Unfortunately, it is **impossible** to learn the distribution of individual treatment effects. No observation of Y_0 and Y_1 for the same person.

The simple idea is to consider the marginal distributions, so that the population parameters of interest are:

- $ATE = \mathbb{E}(Y_1 - Y_0)$
- $ATT = \mathbb{E}(Y_1 - Y_0 | D = 1)$
- $ATNT = \mathbb{E}(Y_1 - Y_0 | D = 0)$

We will encounter more parameters of interest.

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Bounds: What can we learn without further assumptions?

(Based on Slides by Francis DiTraglia)

Manski has pioneered an approach to understand the effect of the assumptions we impose on the data.

The main point was to ask: ‘What can be learnt without additional assumptions? Which values of the parameter of interest can we plausibly exclude?’

Then, how do the additional assumptions we make contribute to excluding additional values?

This is sometimes called Partial Identification.

Bounding the ATE when Y and D are Binary

Example:

$$Y = 1 \text{ if you earn a PhD}, \quad D = 1 \text{ if you attend an Ivy League University.}$$

We know that D is *not* randomly assigned, so we expect selection bias.

Starting point:

- Assume (Y, D) are observed.
- Since Y is binary, $-1 \leq \text{ATE} \leq 1$ even without data.

$$0 \leq Y_0 \leq 1, \quad 0 \leq Y_1 \leq 1 \Rightarrow 0 \leq \mathbb{E}(Y_0) \leq 1, \quad 0 \leq \mathbb{E}(Y_1) \leq 1.$$

Assumption-Free Bounds: Improving on $-1 \leq \text{ATE} \leq 1$

Y and D are observed:

$$P_{11} = \mathbb{E}[Y_1|D=1], \quad P_{10} = \mathbb{E}[Y_0|D=0], \quad p = \mathbb{E}(D).$$

Iterated expectations:

$$\mathbb{E}[Y_1] = \mathbb{E}_D[\mathbb{E}(Y_1|D)] = P_{11}p + \mathbb{E}[Y_1|D=0](1-p),$$

$$\mathbb{E}[Y_0] = \mathbb{E}_D[\mathbb{E}(Y_0|D)] = \mathbb{E}[Y_0|D=1]p + P_{10}(1-p).$$

Bound the unobserved quantities:

$$\mathbb{E}[Y_1|D=0],\ \mathbb{E}[Y_0|D=1] \in [0,1].$$

$$pP_{11} \leq \mathbb{E}[Y_1] \leq pP_{11} + (1-p),$$

$$(1-p)P_{10} \leq \mathbb{E}[Y_0] \leq p + (1-p)P_{10}.$$

Assumption-Free Bounds: Width Equals 1

From previous slide:

$$pP_{11} \leq \mathbb{E}[Y_1] \leq pP_{11} + (1-p),$$

$$(1-p)P_{10} \leq \mathbb{E}[Y_0] \leq p + (1-p)P_{10}.$$

Combine these:

$$pP_{11} - (1-p)P_{10} - p \leq \mathbb{E}[Y_1 - Y_0] \leq pP_{11} - (1-p)P_{10} + (1-p).$$

Written compactly:

$$q \leq \text{ATE} \leq (q+1), \quad q \equiv [pP_{11} - (1-p)P_{10} - p].$$

Half as wide as $[-1, 1]$, but always includes zero.

Add Assumptions to Tighten the Bounds

Monotone Treatment Selection (MTS)

- Suppose direction of self-selection is known; e.g. positive:

$$\mathbb{E}(Y_1|D=0) \leq \mathbb{E}(Y_1|D=1), \quad \mathbb{E}(Y_0|D=0) \leq \mathbb{E}(Y_0|D=1).$$

- Positive MTS improves the **upper bound**:

$$q \leq \text{ATE} \leq P_{11} - P_{10} \leq (q+1).$$

Monotone Treatment Response (MTR)

- Suppose we know direction of causal effect, e.g. $Y_1 > Y_0$.
- Positive MTR gives an improved **lower bound**:

$$0 \leq \text{ATE} \leq (q+1).$$

A Comparison of Bounds

- The preceding bounds are **sharp** under their respective assumptions.
- Example: 8% of Ivy League graduates earn a PhD vs. 1.5% of the general public, and 0.2% attend an Ivy League institution.

$$(P_{11} = 0.08,\ P_{10} = 0.015,\ p = 0.002) \Rightarrow q = [pP_{11} - (1-p)P_{10} - p] \approx -0.017.$$

Bounds under different assumptions:

$$\begin{aligned}
\text{No assumptions:} \quad & [q, q+1] \approx [-0.017, 0.983], \\
\text{Positive MTS:} \quad & [q, P_{11} - P_{10}] \approx [-0.017, 0.065], \\
\text{Positive MTR:} \quad & [0, q+1] \approx [0, 0.983], \\
\text{Positive MTS + MTR:} \quad & [0, P_{11} - P_{10}] = [0, 0.065].
\end{aligned}$$

Here, positive MTR has little effect; positive MTS makes a dramatic difference!

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The Stable Unit Treatment Value Assumption (SUTVA)

In words

A person's outcome depends only on her own treatment.

In math

$$Y_i = (1 - D_i)Y_{i0} + D_i Y_{i1} = Y_{i0} + D_i(Y_{i1} - Y_{i0}).$$

Spillovers (aka interference)

A person's outcome depends on the treatments of other people.

Terminology

Manski calls SUTVA *individualistic treatment response (ITR)*.

Spillovers

Two Kinds of Causal Effects:

- **Direct:** causal effect of Alice's treatment on her own outcome.
- **Indirect:** causal effect of Alice's treatment on Bob's outcome.

Note: Some people say there is a third effect (total effect) — this is just the sum of direct and indirect effects.

Empirical Example with Potential for Indirect Effects

Crépon et al. (2013; QJE)

The experiment

- Large-scale job-seeker assistance program in France.
- Randomized offers of intensive job placement services.

Treatment effects

- *Direct:* If Alice receives job placement, will she be more likely to find a job?
- *Indirect:* If Alice receives job placement, will Bob be less likely to find a job?

Policy question: Do indirect effects partially or completely offset direct effects?

Why might we expect indirect effects?

Crépon et al. (2013) — Displacement effects of labor market policies

"Job seekers who benefit from counseling may be more likely to get a job, but at the expense of other unemployed workers with whom they compete in the labor market. This may be particularly true in the short run, during which vacancies do not adjust: the unemployed who do not benefit from the program could be partially crowded out."

Potential Outcomes with Spillovers / Interference

Notation

- $D = (D_1, D_2, \ldots, D_N)$: vector of treatments of everyone in the experiment.
- Use parentheses rather than subscripts: fully general potential outcomes $Y_i(D)$.

If everyone's treatment can affect Alice, she has 2^N potential outcomes (one for each allocation) — clearly infeasible for inference without structure.

Partial Interference

Problem: Too many potential outcomes.

Solution (Partial interference): assume only a subset of other people affect Alice.

- Suppose there are G groups indexed $g = 1, \ldots, G$.
- Each person belongs to a single, known group; denote person i in group g by (i, g).
- D_g is the vector of treatments for group g.
- **Partial Interference:** $Y_{ig}(D) = Y_{ig}(D_g)$ — spillovers within groups but not between groups.

(Manski calls this "interactions in a reference group.")

Anonymous Interactions

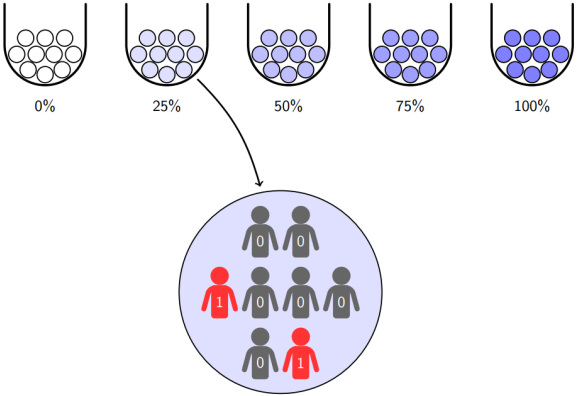
Let N_g be number of people in group g.

- Partial interference reduces potential outcomes from 2^N to 2^{N_g}.
- Still large; impose **anonymous interactions**: only the *number* (or share) of treated neighbors matters, not the identities.
- Let $\bar{D}_{ig}$ denote the share of i's neighbors who are treated.
- Potential outcomes become $Y_{ig}(D) = Y_{ig}(D_{ig}, \bar{D}_{ig})$.

The Randomized Saturation Design

Two-stage experiment for studying spillovers under partial interference (Hudgens & Halloran 2008):

- Randomly assign *saturations* S_g to groups (the share of people treated).
- Given S_g, randomly assign treatments to individuals according to that group's saturation.



Identifying the Direct Effect

Notation

- Y_{ig}: observed outcome of person (i, g).
- $S_g \in \mathscr{S} \subseteq [0, 1]$: saturation assigned to group g.
- $D_{ig} \in \{0, 1\}$: treatment assigned to person (i, g).
- Random assignment: S_g and D_{ig} independent of potential outcomes.

Direct effect at saturation s:

$$DE(s) = E[Y_{ig} \mid D_{ig} = 1, S_g = s] - E[Y_{ig} \mid D_{ig} = 0, S_g = s] = E[Y_{ig}(1, s) - Y_{ig}(0, s)].$$

Identifying the Indirect Effects

Indirect effect when untreated: $IE_0(s \to s')$

$$IE_0(s \to s') = E[Y_{ig} \mid D_{ig} = 0, S_g = s'] - E[Y_{ig} \mid D_{ig} = 0, S_g = s] = E[Y_{ig}(0, s') - Y_{ig}(0, s)].$$

Indirect effect when treated: $IE_1(s \to s')$

$$IE_1(s \to s') = E[Y_{ig} \mid D_{ig} = 1, S_g = s'] - E[Y_{ig} \mid D_{ig} = 1, S_g = s] = E[Y_{ig}(1, s') - Y_{ig}(1, s)].$$

These effects are 'intent-to-treat'. They do not consider non-compliance. For this case, see DiTraglia et al. (2023)

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The "credibility revolution" (Angrist and Pischke, JEP 2010)

"Hardly anyone takes data analysis seriously. Or perhaps more accurately, hardly anyone takes anyone else's data analysis seriously." Leamer (1983) quoted from Angrist and Pischke (JEP, 2010, henceforth AP)

"Empirical microeconomics has experienced a ***credibility revolution****, with a consequent increase in policy relevance and scientific impact.[...]*

Sensitivity analysis played a role in this, but as we see it, the primary engine driving improvement has been a focus on the ***quality of empirical research designs****."* AP

The "credibility revolution" (Angrist and Pischke, JEP 2010)

Some examples of influential design-based research:

PROGRESA program in Mexico:

- Cash transfer to randomly selected mothers,
- Conditional on participation in prenatal care, nutritional monitoring of children, and children's regular school attendance.

"Progresa is why now thirty countries worldwide have conditional cash transfer programs."
Gertler quoted from AP

The "credibility revolution" (Angrist and Pischke, JEP 2010)

Some examples of influential design-based research:

Fehr and Goette (AER, 2007)'s randomized experiment:

- Measure intertemporal substitution elasticity (response to transitory wage changes),
- randomized the pay of bicycle messengers, offering one group and then another a temporarily higher wage,
- implied intertemporal substitution elasticity of about unity.

The "credibility revolution" (Angrist and Pischke, JEP 2010)

Increased Credibility because causal interpretation does not come from the econometric methodology alone.

Lalonde (AER, 1986):

- Compare the causal effect of a training program on wages from a randomized trial to an econometric evaluation steming from observational data.
- *"Lalonde argued that there is little reason to believe that statistical comparisons of alternative models (specification testing) would point a researcher in the right direction"* AP

The "credibility revolution" (Angrist and Pischke, JEP 2010)

Regression Discontinuity provides a good example of the transparency of the methodology.

Angrist and Lavy (QJE, 1999)

- Effect of Class size on student output;
- Discontinuity: class size in Israel is capped at 40;
- A cohort of 41 is split in two classes;
- Find an increase in achievement with smaller class sizes.

The "credibility revolution" (Angrist and Pischke, JEP 2010)

Regression Discontinuity rests on assumptions that are fairly easy to assess.

Angrist and Lavy (QJE, 1999)

- Possibility to test the random assignment around the threshold:
- Using successively narrower windows;
- Looking at bunching in observed characteristics.

The "credibility revolution" (Angrist and Pischke, JEP 2010)

Advantages of a thorough research design

1. Simple explanation : difference between control and treatment.
2. Narrower specification analysis:
   - RCT: balance of pre-treatment characteristics;
   - IV: Existence of other channels through which IV influence the outcome;
   - RDD: Look for bunching in observed covariates;
   - Diff-in-Diff : Examine parallel trends.

The "credibility revolution" (Angrist and Pischke, JEP 2010)

Addressing Critiques : External validity

"The criticism here [...] is that in the quest for internal validity, design-based studies have become narrow or idiosyncratic." AP

"Empirical evidence on any given causal effect is always local, derived from a particular time, place, and research design. Invocation of a superficially general structural framework does not make the underlying variation or setting more representative.[...] [E]xtrapolation of causal effects to new settings is always speculative." AP, my own emphasis.

The "credibility revolution" (Angrist and Pischke, JEP 2010)

Addressing Critiques: External Validity

"A constructive response to the specificity of a given research design is to look for more evidence, so that a more general picture begins to emerge." AP

"Reasonably well-identified studies from a number of advanced countries, at different grade levels and subjects, and for class sizes ranging anywhere from a few students to about 40, have produced estimates [...] a ten-student reduction in class size produces about a 0.2 to 0.3 standard deviation increase in individual test scores."

A structural perspective on the experimentalist school (Keane, JEP 2010)

"A False panacea"

"What has always bothered me about the 'experimentalist' school is the false sense of certainty it conveys. The basic idea is that if we have a 'really good instrument' we can come up with 'convincing' estimates of 'causal effects' that are not 'too sensitive to assumptions.'" K

A structural perspective on the experimentalist school (Keane, JEP 2010)

Example of Angrist and Lavy (1999)

Threats to internal validity: what if ...

- Schools that perform well attract more students?
- Parents who care more respond to large class size by sending there children to different schools?
- Teachers assigned to extra classes are not a random samples of teachers?

A structural perspective on the experimentalist school (Keane, JEP 2010)

A more fundamental problem: credibility of the causality

A number of unobserved:

- "quality" time with parents;
- books at home, educational toys;
- nutrition, time spent watching TV, etc.

"... a quasi-experimental design is not a ceteris paribus effect. It subsumes how class size influences all the factors we haven't controlled, including parent/school reactions to mandated class size changes." K

A structural perspective on the experimentalist school (Keane, JEP 2010)

The key problem: "So what?"

We don't have:

- structural parameters of the cognitive ability production function,
- decision rule of parents and schools;

"... we cannot determine if reduced class size would be a more cost effective way to improve student achievement than, say, higher teacher salaries or better nutrition and health care in utero." K

A structural perspective on the experimentalist school (Keane, JEP 2010)

We can't escape assumptions:

"... what bothers me is not the natural experiment approach per se, but rather the exaggerated claim that it enables us to attain relatively assumption-free statistical inference."

"I'm not dismissing the Angrist and Lavy (1999) result. I take it as one piece of evidence [...]. But I would not take it as definitive." K

A structural perspective on the experimentalist school (Keane, JEP 2010)

In marketing, experiments have proven of limited use... Where as the structural paradigm is dominant.

How demand vary with a price cut is not very interesting... Rather:

- of increase in sale, how much comes from stealing vs. category expansion vs. cannibalization?
- how much price cuts reduce brand equity?
- What would be profits under every-day low price vs. policy of frequent price cuts?

A structural perspective on the experimentalist school (Keane, JEP 2010)

Answers from Erdem, Imai and Keane (2003) and Erdem, Keane and Sun (2008):

Elasticity to price 3 to 4.5:

- 20-30% cannibalization;
- of the remaining, 70-80% category expansion;
- 20-30% brand switching.

A structural perspective on the experimentalist school (Keane, JEP 2010)

Good data always help:

"Plausibly exogenous variation in variables of interest is a desideratum in all empirical work - not an argument for one approach over another."

"I agree that all econometric work, whether structural or not, should ideally be based on such plausibly exogenous variation in the data." K

A structural perspective on the experimentalist school (Keane, JEP 2010)

"Where I most strongly disagree with Angrist and Pischke is their notion that empirical work can exist independently from, or occur prior to, economic theory."

"Yet, as should be obvious, a scientist doesn't do experiments or run clinical trials as 'accidental play, without pre-existence of more or less definite ideas about their meaning' (Einstein and Infeld, 1938)." K

A structural perspective on the experimentalist school (Keane, JEP 2010)

To conclude:

"all inferences are based on some ultimately untestable assumptions that must, therefore, be made based on the "whim" (or opinion) of the investigator."

- we should report on sensitivity to specification,
- and remember that no instrument will be convincing to everyone.

A structural perspective on the experimentalist school (Keane, JEP 2010)

External validation in structural approach

- Keane and Moffit (1998): Forecast behavior prior to significant change;
- Keane and Wolpin (2007): Predict behaviour in a holdout state with a different policy regime

"We know perfectly well that our models aren't true. Validation exercises are used purely as a way to persuade the audience (and ourselves) that a model may be a useful tool for prediction and policy evaluation."

Taking stocks

What do you think?

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